



DEALER BEST PRACTICE SERIES

Series High Efficiency Fuel
Filters

Application Maintenance Site Management Component Rebuild **Component Life Management** Safety MARC Management

Series High Efficiency Fuel Filters0

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1.0 Introduction

Modern high-pressure, electronically controlled fuel systems are used on nearly all Caterpillar engines. These systems deliver better performance and improved fuel economy over older mechanical systems. In general, electronic injectors should last until engine overhaul if relatively clean fuel is used along with high quality, high efficiency fuel filters.

However, because modern injectors produce higher injection pressures, (typically 25,000 – 30,000 psi) they are subject to accelerated wear from abrasive particles in the fuel. Using fuel with high levels of particulate contaminants or continuing to operate the machine after the plugged fuel filter light has been activated can cause rapid injector wear.



New and damaged injector poppet valves. Abrasive debris flowing at very high velocity across the poppet seat causes erosion damage and internal injector leakage.

Adding an additional fuel filter in series with the standard fuel filter can significantly improve injector life.

2.0 Best Practice Description

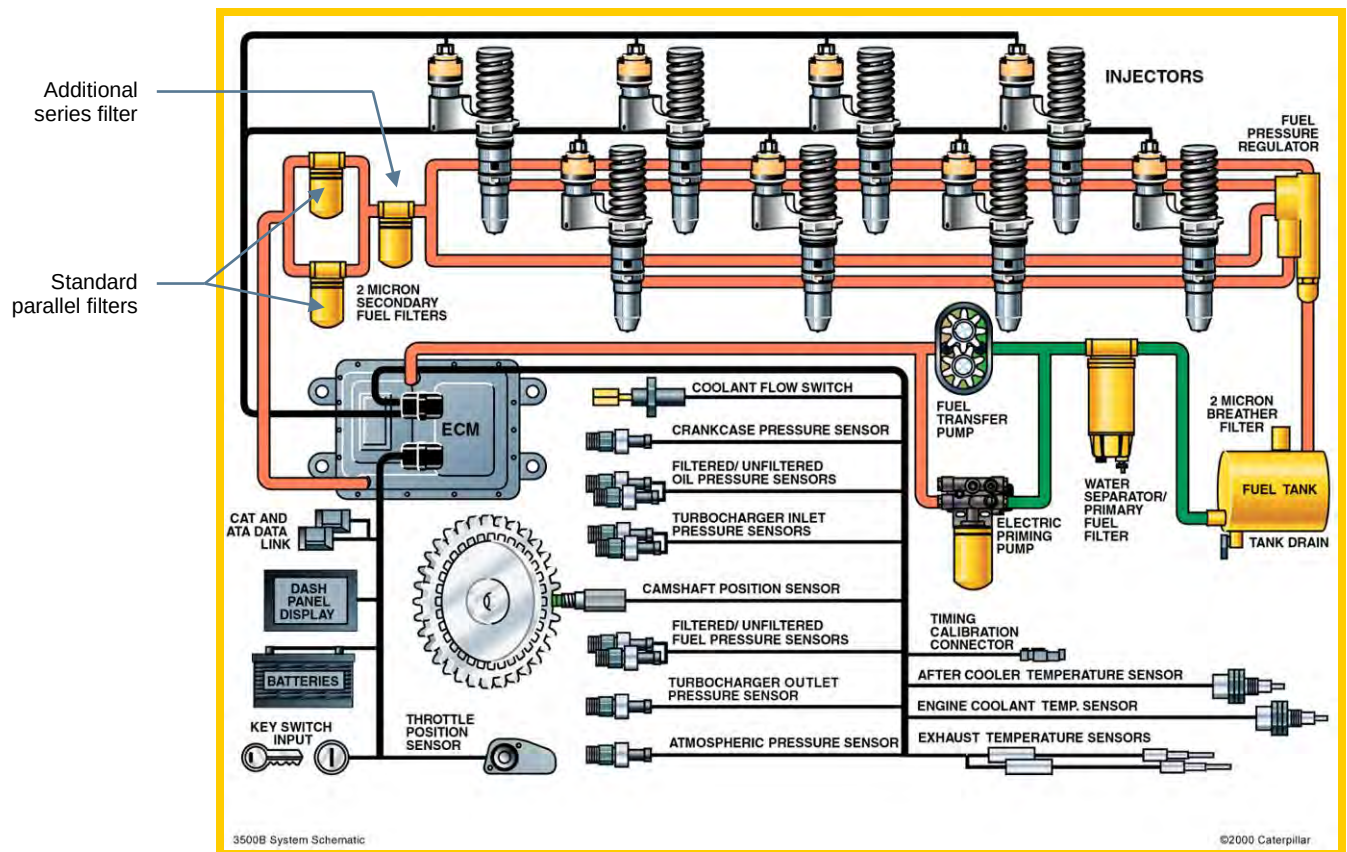
3500 Series engines use two 4-micron fuel filters in parallel to provide adequate debris holding capacity. This arrangement provides single pass filtration, which is adequate if:

- The fuel is reasonably clean
- Fuel filters are changed before or immediately after the plugged filter light comes on

The filter media (paper) in the filter, which actually traps and holds the particulate debris, is capable of single pass efficiency in excess of 98%. However, as the debris holding capacity of the media is reached, flow restriction through the media increases and the fuel pressure drop across the media starts to increase. This pressure difference between the inlet and outlet sides of the filter is commonly referred to as “delta p.” Filter efficiency typically remains high until the filter reaches about 18 – 20 psi delta p. At this point, the pressure differential exceeds the ability of the media to hold trapped debris particles and begins to push debris through the filter media. As delta p increases, the size and number of particles pushed through the media increases sharply. These particles cause injector abrasive wear and loss of performance.

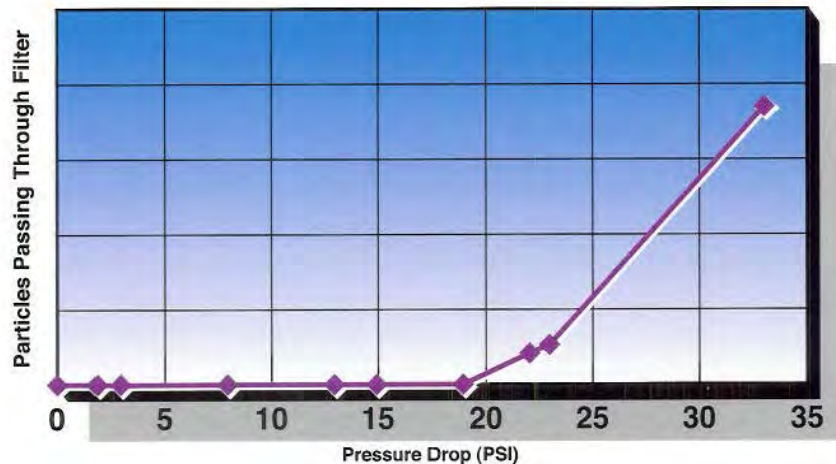
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Typical 3500 Series Fuel System with Series Filter Added

ABRASIVE PARTICLES FORCED THROUGH SECONDARY FUEL FILTER



Below 18 psi pressure drop, high efficiency fuel filters capture and retain nearly 100% of abrasive particles. Pressure drops in excess of 18 psi force debris particles through the filter media and downstream. As restriction increases, the size and number of particles forced through the filter media increases dramatically.

Adding an additional (series) filter downstream of the factory parallel filters traps particles, which may otherwise be pushed through the media. Most 3500 engines will operate without noticeable

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performance loss to as low as 10 psi fuel pressure to the injectors. With a nominal fuel supply pressure to the fuel filter of 65 psi, it is possible to achieve delta p in excess of 50 psi if the filter becomes severely plugged. This will allow very large amounts of debris to reach the injectors if a series filter is not used. Keep in mind that a series filter will not cause additional plugging, it will simply catch debris pushed through the first filter.

3.0 Implementation Steps

Factory arrangements are not available. However, the series filter can be added by fabricating a mounting bracket and hoses.



Series filter (center) installed downstream of the factory parallel filters (left)

4.0 Benefits

Installing series filtration can protect the injectors and help them last until engine overhaul. It is not uncommon for applications with poor fuel quality to replace injectors at engine mid-life. On a 3516, the cost of mid-life injector replacement is \$20,000 or more. Using series filtration can often eliminate this cost.

5.0 Resources Required

- Additional High Efficiency filter and filter base
 - o Filter P/N 1R-0755 for 3500 Series engines
- Fabricated filter mounting bracket and hoses

6.0 Supporting Attachments / References

Improving Component Durability booklets
Specifically:

Fluid Cleanliness Management

SEBF1020



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7.0 Related Best Practices

None.

8.0 Acknowledgements

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